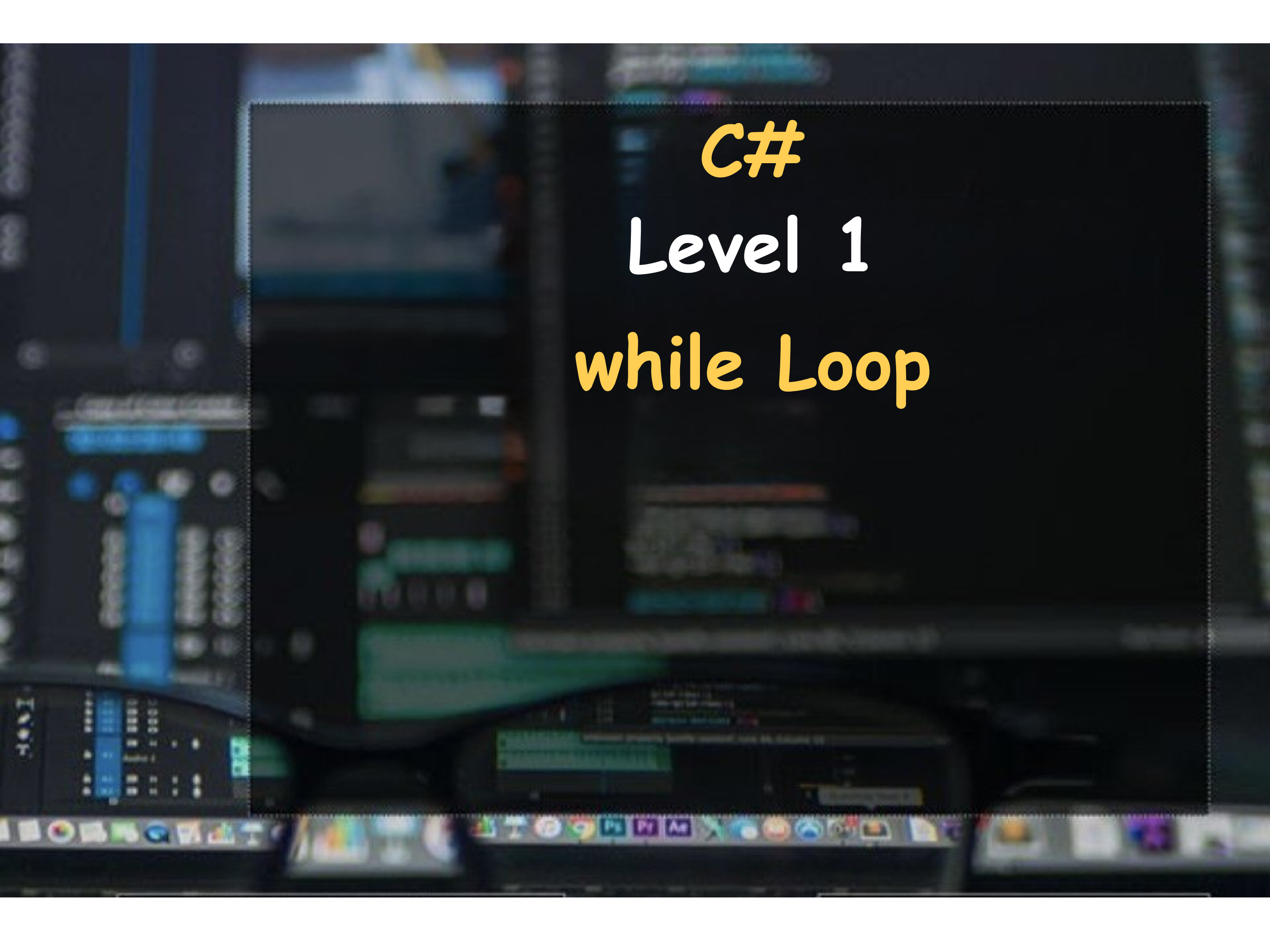
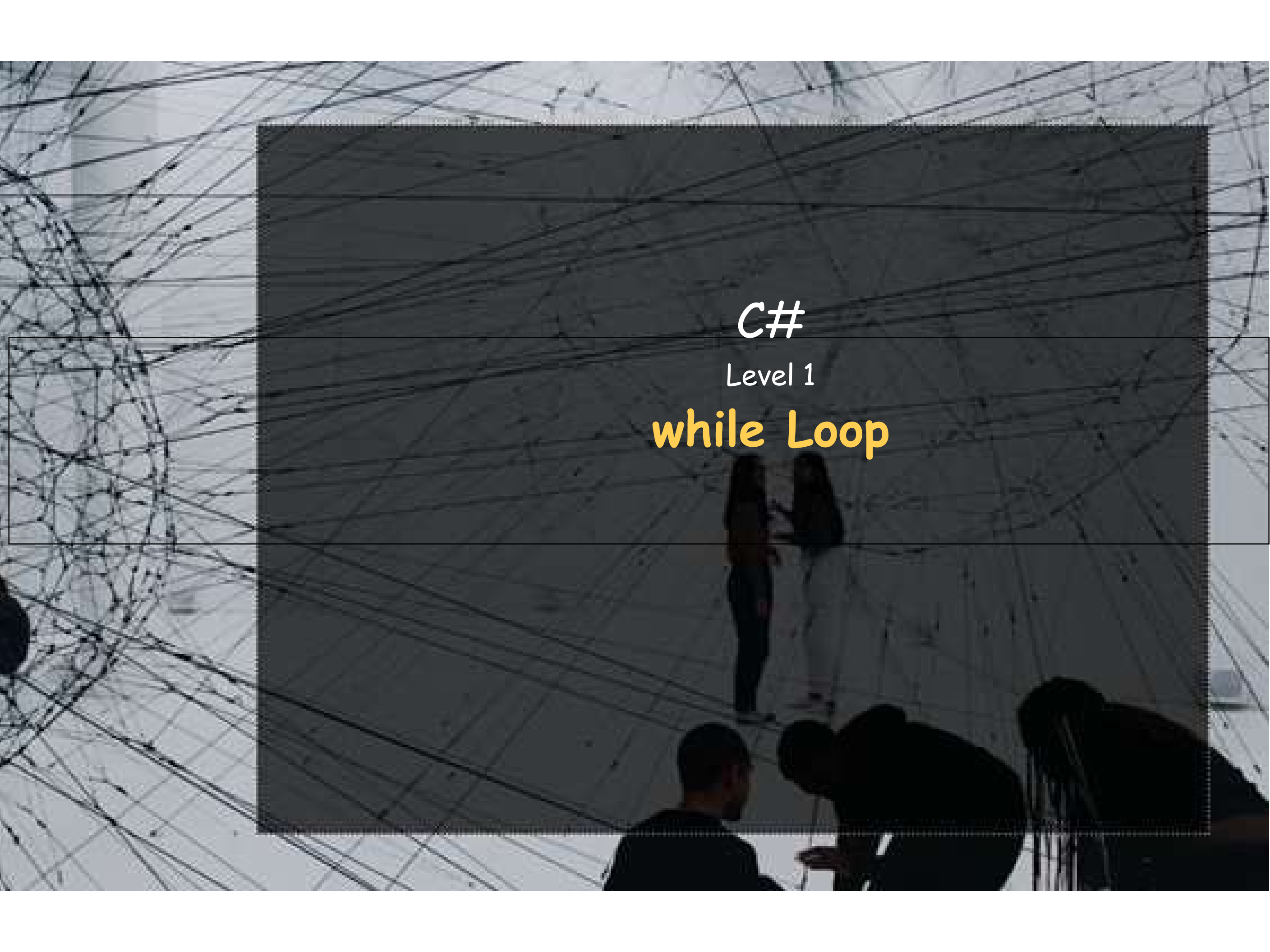




C#

Level 1

while Loop



C#

Level 1

while Loop

Definition

while Loop

-  Repeats a block of code as long as a condition is true.

-  The condition is checked before each iteration.

Procedures

1. Check the condition.
2. If true → execute the loop body.
3. Update the variable.
4. Check the condition again.
5. Stop when the condition becomes false.

💡 Example

```
int i = 1;
while (i <= 5)
{
    Console.WriteLine(i);
    i++;
}
```

➡ Result

```
C# while Loop: Iteration 1
C# while Loop: Iteration 2
C# while Loop: Iteration 3
C# while Loop: Iteration 4
C# while Loop: Iteration 5
```

Note

- ◆ The loop variable must be updated so the condition can eventually become false.
- ◆ Forgetting the update can cause an infinite loop.

Comparison

 for

- ✓ Initialization, condition, and iterator are written together.

 while

- ✓ Initialization and update are usually written separately from the condition.
- ✓ Useful when the number of iterations is not known in advance.

✦✦ Summary

- ✓ while repeats code while a condition is true.
- ✓ The condition is checked before each iteration.
- ✓ The loop stops when the condition becomes false.
- ✓ The variable should be updated to avoid an infinite loop.

Thank you 😊